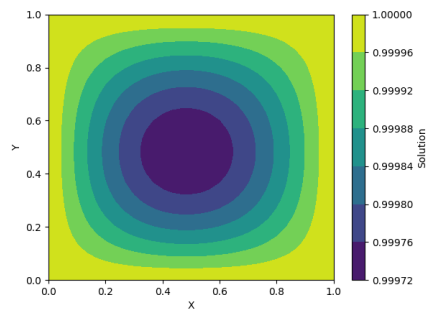

Figure 1: $RHS = 0$, boundary$= 1$

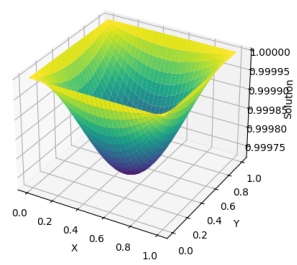

Figure 2: $RHS = 0$, boundary$= 1$

1 Řešení Poissonovy rovnice

Mějme rovnici

$$-\nabla \cdot (p(x,y)\nabla u(x,y)) = f(x,y) \quad \text{na } \Omega = (0,1) \times (0,1). \tag{1}$$

$$f(x,y) = 10,\ p(x,y) = 1,\ \forall (x,y) \in \Omega \tag{2}$$

2 Gauss-Seidel

Při 1000 iteraci jsme dosáhli L2 přesnosti okolo 10^{-5}. Zároveň si můžeme všimnout, že okolo středu se zužuje peak řešení při změně parametrů. Ta byla prováděna pouze pro tuto metodu.

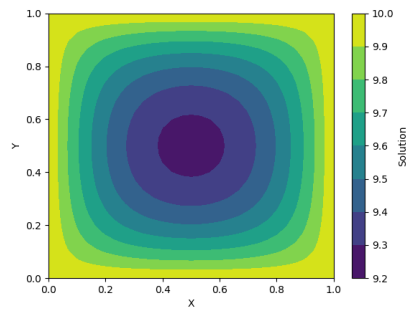

Figure 3: $RHS = 10$, boundary= 10

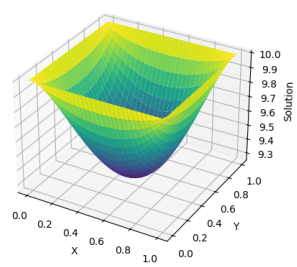

Figure 4: $RHS = 10$, boundary= 10

3 SOR

Vypadá to, že pro SOR metodu je ideální parametr kolem 1.87 viz tabulky s chybou.

4 Jacobi

Pro Jacobiho metodu chyba diverguje.

5 Richardson

Pro Richardsonovu metodu chyba také diverguje.

6 Porovnání

Je vidět že nejvhodnější metoda je metoda SOR s vhodným parametrem.

7 Zmenšování h

Vidíme, že se řešení pomalu zlepšuje za cenu většího počtu iterací, které nejsou dramatické (cca 2× menší h dá 20 iterací navíc).

Table 1: Konvergence L2-rezidua v závislosti na počtu iterací GS

Iterace	L_2-reziduum	Iterace	L_2-reziduum
0	85.685591	500	0.020959
10	9.952980	510	0.018778
20	5.913682	520	0.016824
30	4.356655	530	0.015074
40	3.534928	540	0.013506
50	3.020732	550	0.012100
60	2.649950	560	0.010841
70	2.353448	570	0.009713
80	2.101636	580	0.008703
90	1.881250	590	0.007797
100	1.685680	600	0.006986
110	1.511058	610	0.006259
120	1.354722	620	0.005608
130	1.214592	630	0.005025
140	1.088929	640	0.004502
150	0.976219	650	0.004033
160	0.875125	660	0.003614
170	0.784453	670	0.003238
180	0.703136	680	0.002901
190	0.630214	690	0.002599
200	0.564828	700	0.002329
210	0.506204	710	0.002086
220	0.453647	720	0.001869
230	0.406532	730	0.001675
240	0.364301	740	0.001501
250	0.326448	750	0.001344
260	0.292521	760	0.001205
270	0.262116	770	0.001079
280	0.234867	780	0.000967
290	0.210447	790	0.000866
300	0.188565	800	0.000776
310	0.168956	810	0.000695
320	0.151385	820	0.000623
330	0.135640	830	0.000558
340	0.121532	840	0.000500
350	0.108890	850	0.000448
360	0.097564	860	0.000402
370	0.087415	870	0.000360
380	0.078321	880	0.000322
390	0.070174	890	0.000289
400	0.062873	900	0.000259
410	0.056332	910	0.000232
420	0.050472	920	0.000208
430	0.045221	930	0.000186
440	0.040516	940	0.000167
450	0.036301	950	0.000149
460	0.032524	960	0.000134
470	0.029141	970	0.000120
480	0.026109	980	0.000107
490	0.023392	990	9.6257e-05

Table 2: Konvergence L2-rezidua (rychlá konvergence) SOR parametr: 1.87

Iterace	L_2-reziduum
0	150.709 633 925 8
10	224.302 924 578 6
20	119.136 198 130 3
30	60.428 155 960 0
40	30.176 874 774 9
50	14.552 113 948 3
60	5.627 645 099 4
70	1.403 994 231 9
80	0.351 554 466 1
90	0.089 989 805 9
100	0.024 817 661 8
110	0.008 004 149 6
120	0.002 636 838 2
130	0.000 658 712 6
140	0.000 164 287 3
150	4.1354e-05

Table 3: Konvergence L2-rezidua v závislosti na počtu iterací SOR parametr: 1

Iterace	L_2-reziduum	Iterace	L_2-reziduum
0	85.685591	500	0.020959
10	9.952980	510	0.018778
20	5.913682	520	0.016824
30	4.356655	530	0.015074
40	3.534928	540	0.013506
50	3.020732	550	0.012100
60	2.649950	560	0.010841
70	2.353448	570	0.009713
80	2.101636	580	0.008703
90	1.881250	590	0.007797
100	1.685680	600	0.006986
110	1.511058	610	0.006259
120	1.354722	620	0.005608
130	1.214592	630	0.005025
140	1.088929	640	0.004502
150	0.976219	650	0.004033
160	0.875125	660	0.003614
170	0.784453	670	0.003238
180	0.703136	680	0.002901
190	0.630214	690	0.002599
200	0.564828	700	0.002329
210	0.506204	710	0.002086
220	0.453647	720	0.001869
230	0.406532	730	0.001675
240	0.364301	740	0.001501
250	0.326448	750	0.001344
260	0.292521	760	0.001205
270	0.262116	770	0.001079
280	0.234867	780	0.000967
290	0.210447	790	0.000866
300	0.188565	800	0.000776
310	0.168956	810	0.000695
320	0.151385	820	0.000623
330	0.135640	830	0.000558
340	0.121532	840	0.000500
350	0.108890	850	0.000448
360	0.097564	860	0.000402
370	0.087415	870	0.000360
380	0.078321	880	0.000322
390	0.070174	890	0.000289
400	0.062873	900	0.000259
410	0.056332	910	0.000232
420	0.050472	920	0.000208
430	0.045221	930	0.000186
440	0.040516	940	0.000167
450	0.036301	950	0.000149
460	0.032524	960	0.000134
470	0.029141	970	0.000120
480	0.026109	980	0.000107
490	0.023392	990	9.6257e-05

Table 4: Konvergence normy L_2-rezidua SOR parametr: 1.5

Iterace	L_2-reziduum
0	127.269 060
10	5.261 414
20	2.665 107
30	1.885 168
40	1.371 028
50	0.997 250
60	0.721 961
70	0.520 196
80	0.373 497
90	0.267 544
100	0.191 371
110	0.136 766
120	0.097 691
130	0.069 759
140	0.049 804
150	0.035 554
160	0.025 379
170	0.018 116
180	0.012 931
190	0.009 230
200	0.006 588
210	0.004 702
220	0.003 356
230	0.002 396
240	0.001 710
250	0.001 221
260	0.000 871
270	0.000 622
280	0.000 444
290	0.000 317
300	0.000 226
310	0.000 161
320	0.000 115
330	8.2223e-05

Table 5: Divergence řešení (nestabilní výpočet) jacobi

Iterace	L_2-reziduum	Iterace	L_2-reziduum
0	175.7183	210	1.6373e+82
10	1.1916e+05	220	1.4428e+86
20	3.0197e+08	230	1.2810e+90
30	1.3708e+12	240	1.1452e+94
40	7.7801e+15	250	1.0298e+98
50	4.9746e+19	260	9.3076e+101
60	3.4285e+23	270	8.4477e+105
70	2.4891e+27	280	7.6939e+109
80	1.8767e+31	290	7.0269e+113
90	1.4544e+35	300	6.4322e+117
100	1.1497e+39	310	5.8983e+121
110	9.2197e+42	320	5.4164e+125
120	7.4747e+46	330	4.9793e+129
130	6.1137e+50	340	4.5813e+133
140	5.0401e+54	350	4.2180e+137
150	4.1867e+58	360	3.8856e+141
160	3.5050e+62	370	3.5807e+145
170	2.9582e+66	380	3.3008e+149
180	2.5181e+70	390	3.0435e+153
190	2.1624e+74	400	**inf**
200	1.8735e+78	780	**nan**

Table 6: Divergence řešení Richardson

Iterace	L_2-reziduum
0	1.1576×10^{28}
10	3.9980×10^{51}
20	1.5370×10^{71}
30	4.3327×10^{89}
40	3.4559×10^{107}
50	1.3434×10^{125}
60	3.3319×10^{142}
70	**inf**
80	**inf**
90	**inf**
100	**inf**
110	**inf**
120	**inf**
130	**inf**

Table 7: Vývoj L_2-rezidua pro iterační proces s menšeným h SOR 1.87 h = 1/29

Iterace	L_2-reziduum
0	150.709 633 925 8
10	224.302 924 578 6
20	119.136 198 130 3
30	60.428 155 960 0
40	30.176 874 774 9
50	14.552 113 948 3
60	5.627 645 099 4
70	1.403 994 231 9
80	0.351 554 466 1
90	0.089 989 805 9
100	0.024 817 661 8
110	0.008 004 149 6
120	0.002 636 838 2
130	0.000 658 712 6
140	0.000 164 287 3
150	4.1354×10^{-5}

Table 8: Vývoj L_2-rezidua pro iterační proces s menšeným h SOR 1.87 h = 1/49

Iterace	L_2-reziduum
0	206.429 839 234 1
10	286.814 917 024 8
20	136.733 459 002 1
30	65.767 142 105 7
40	32.421 801 262 9
50	16.120 657 998 6
60	8.030 010 595 7
70	3.994 888 279 0
80	1.970 901 540 9
90	0.936 263 062 7
100	0.358 621 105 8
110	0.089 508 857 9
120	0.022 278 012 8
130	0.005 559 990 6
140	0.001 400 876 0
150	0.000 363 113 3
160	0.000 101 551 6
170	3.2852×10^{-5}